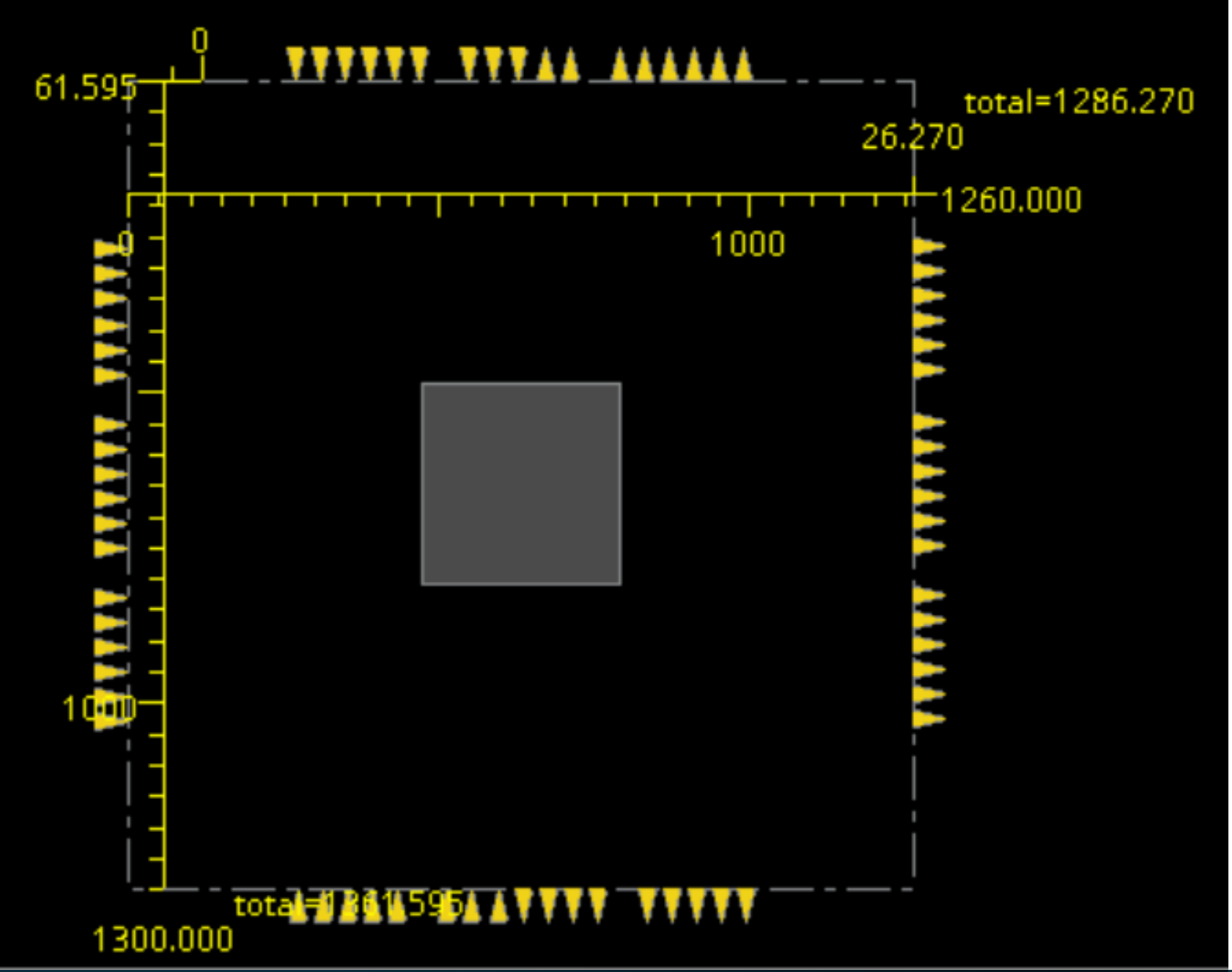
Final project

Cpu_32bit

Nguyễn Thị Mỹ Lĩnh

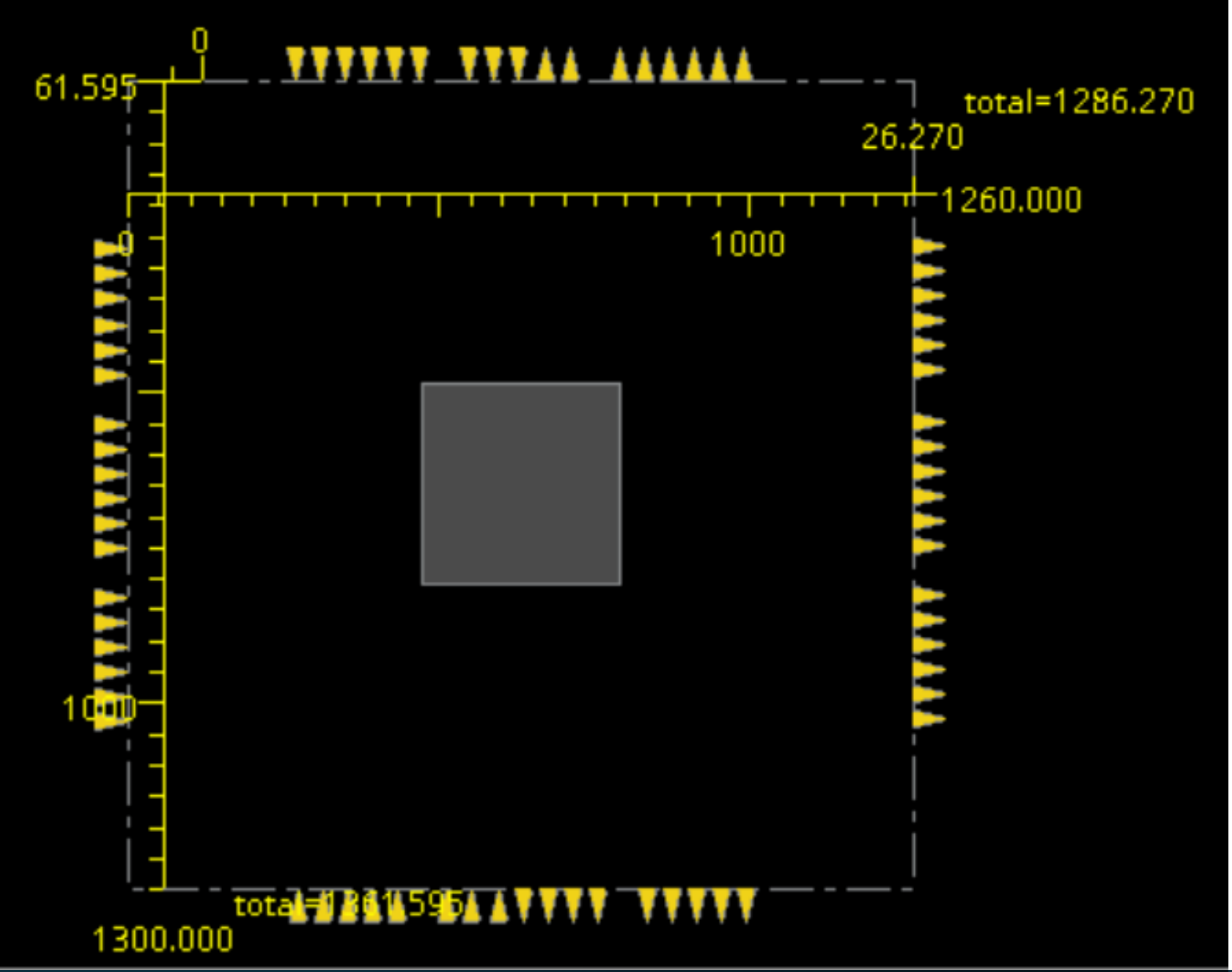


Floorplan

- Die sizing caculation :
- PadPitch : 40
- Corner : 250

- Cpu32bit has 72 signal IO ports (ctrl + 32 din + 32 dout + status)

- Add per-side PG pads (VDD+VSS):
- Npads_top = 17+ 2 = 19
- Npads_bot = 17 + 2 = 19
- Npads_left = 18+ 2 = 20
- Npads_right = 18+ 2 = 20



Floorplan

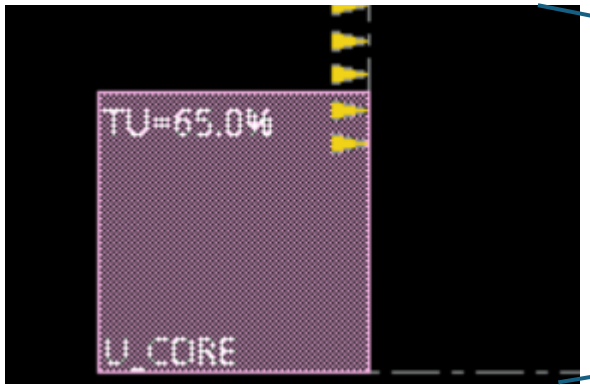
- Die sizing (min):
- dieH = corner(250) + max(Npads_top,Npads_bot)*pitch(40) + corner(250)
- = 250 + max(20,19)*40 + 250
- = 250 + 800 + 250 = 1300

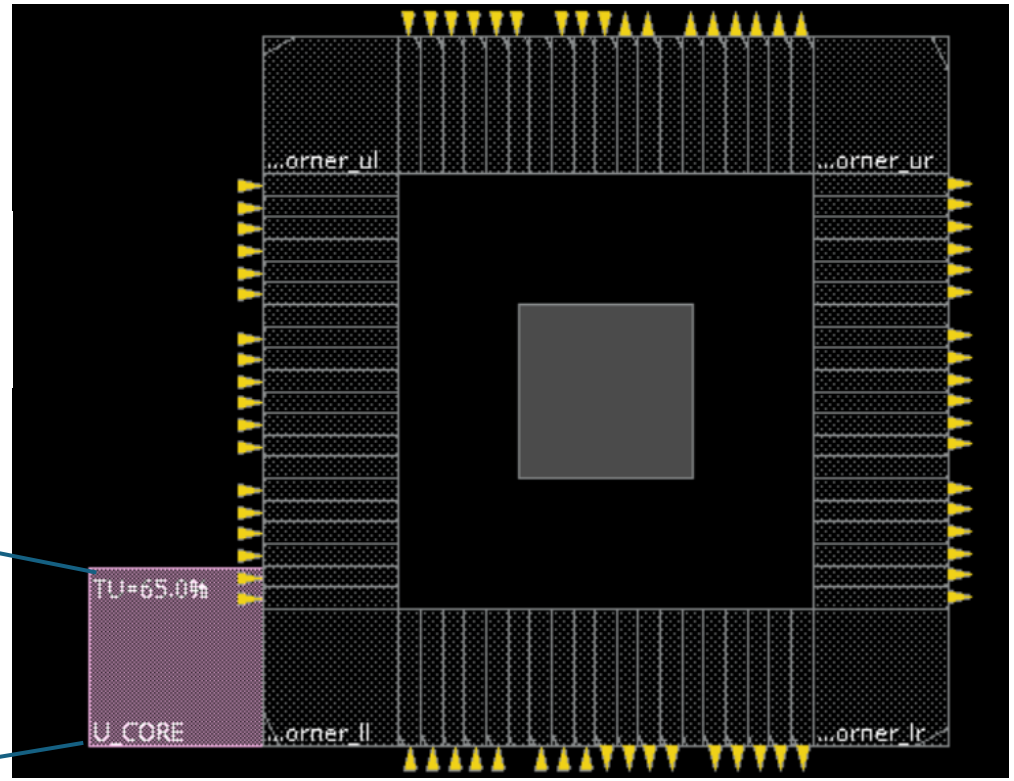
- dieW= corner(250) + max(Npads_left,Npads_right)*pitch(40) + corner(250)
- = 250 + max(19,19)*40 + 250
- = 250 + 760 + 250 = 1260

Floorplan – utilization

- Set uti_target : 0.65
- Logic area from syn output report

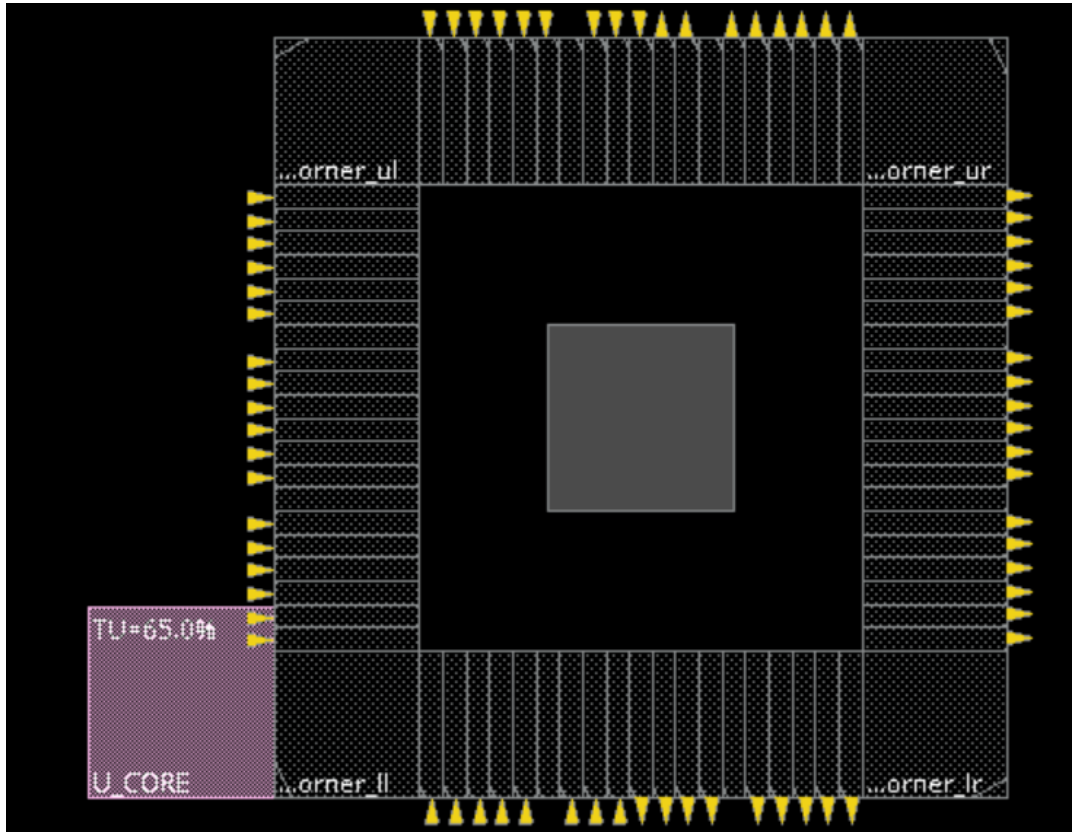
```
set UTIL_TARGET 0.65
set LOGIC_AREA  67155.291
set TARGET_AREA [expr {$LOGIC_AREA / $UTIL_TARGET}]
```





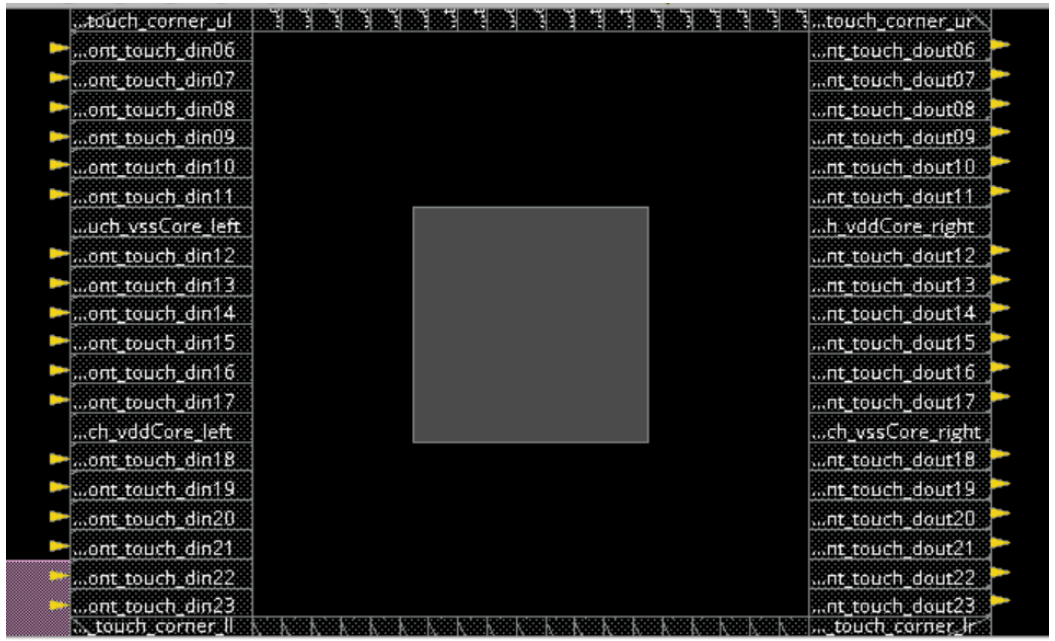
Floorplan – io placement

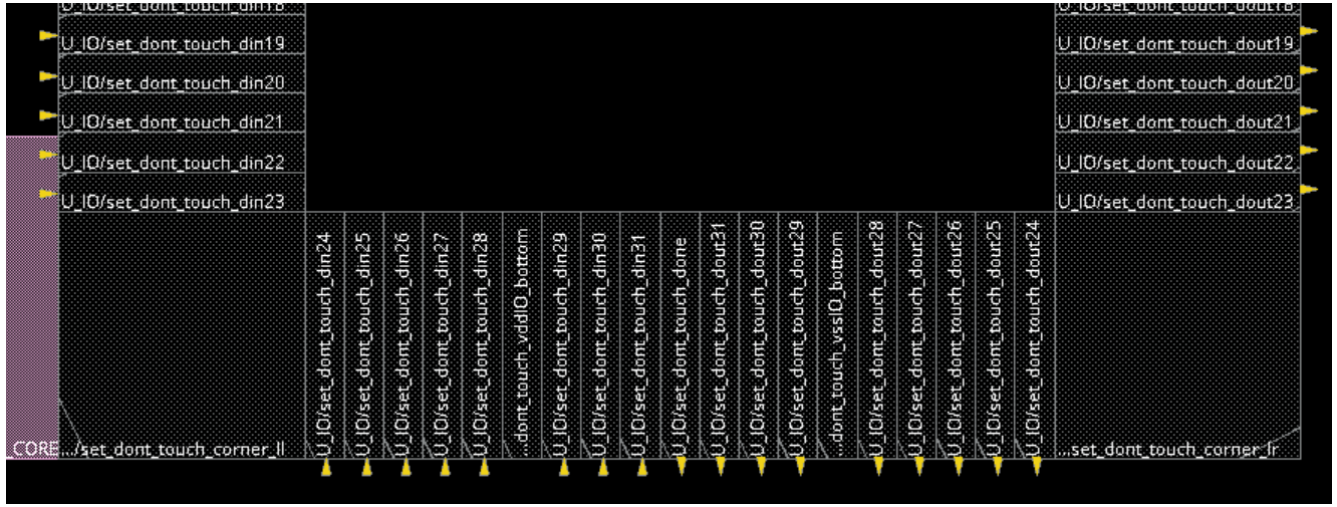
- Left, right : 18 io pad so every 6 io pads place vdd, vss.-> 20 pad
- Top, bot : 19 pad



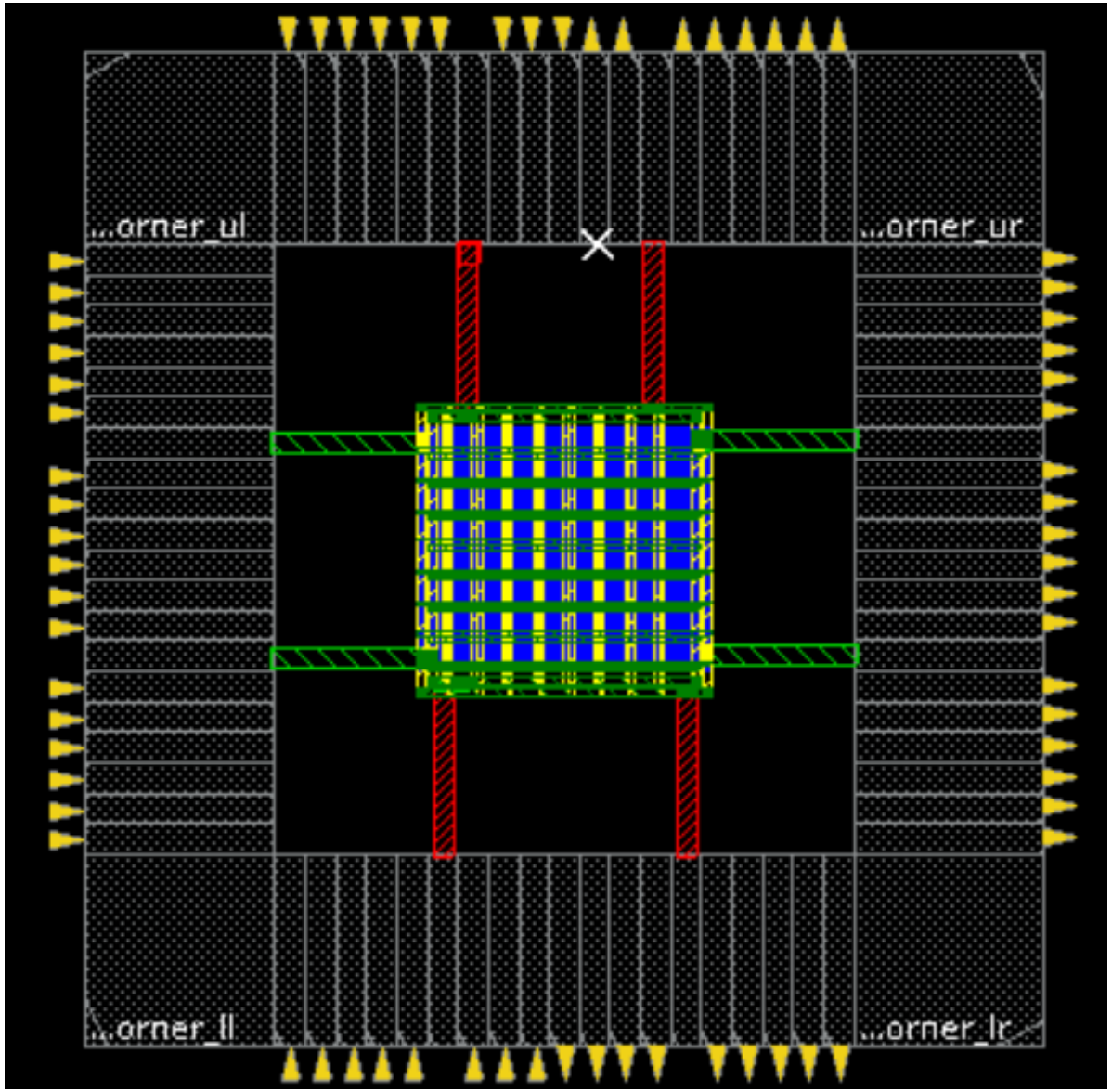
Floorplan – io placement

- Left, right : 18 io pad so every 6 io pads place vdd, vss.-> 20 pad
- Top, bot : 19 pad





Floorplan – power domand



Floorplan – check_floorplan

```
@innovus 35> check_floorplan
Checking routing tracks.....
Checking other grids.....
Checking FINFET Grid is on Manufacture Grid.....
Checking core/die box is on Grid.....
Checking snap rule ......
Checking Row is on grid......
INFO (IMPFP-7241): Tech site CoreSite has no symmetry Y.        Current symmetry setting : isSymmetryR90 false; isSymmetryX : false; isSymme
tryY : false;

Checking AreaIO row.....
Checking row out of die ...
Checking routing blockage.....
Checking placement blockage.....
placeBlkGrid grid_type = 0
Checking components.....
Checking IO Pads out of die...
Checking constraints (guide/region/fence).....

Checking Preroutes.....
No. of regular pre-routes not on tracks : 0
*** Message Summary: 0 warning(s), 0 error(s)

0
@innovus 36>
```

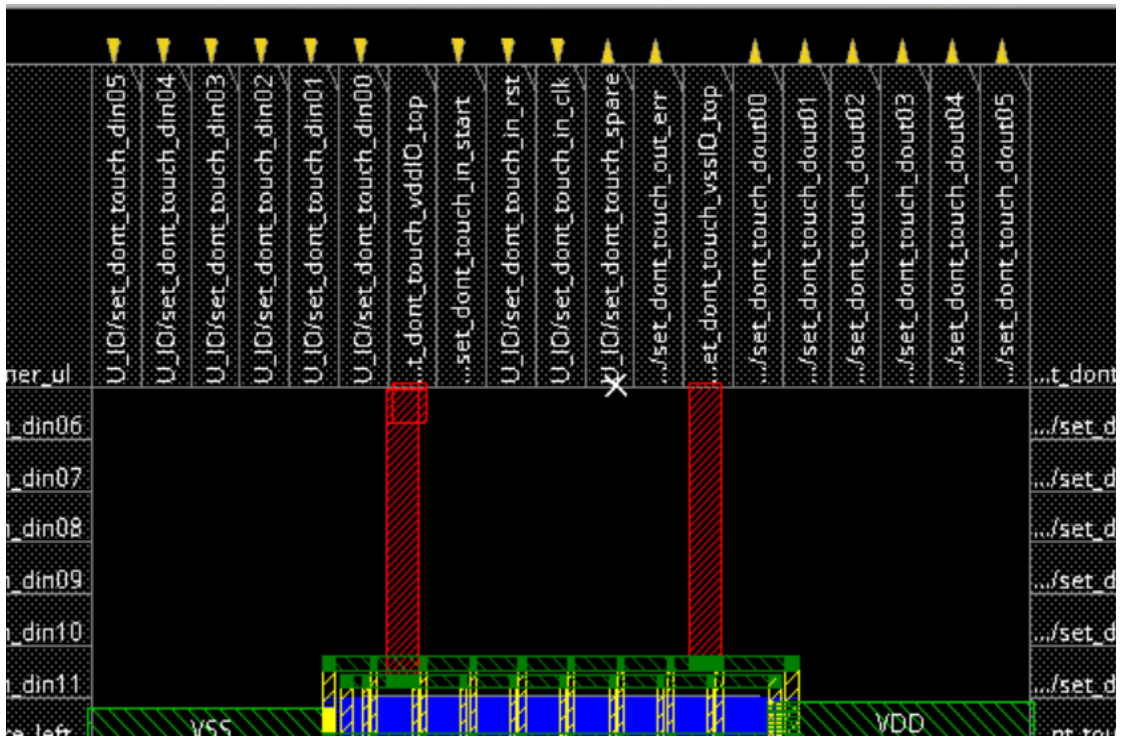
Floorplan – check_floorplan

```
0
@innovus 36> check_place
**Info: (IMPSP-307): Design contains fractional 17 cells.
Begin checking placement ... (start mem=1827.7M, init mem=1957.3M)
*info: Placed = 0
*info: Unplaced = 19337
Placement Density:65.35%(67159/102774)
Placement Density (including fixed std cells):65.35%(67159/102774)
Finished check_place (total: cpu=0:00:00.1, real=0:00:00.0; vio checks: cpu=0:00:00.0, real=0:00:00.0; mem=1957.3M)
0
@innovus 37>
```

```
  VERIFY DRC ...... Sub-Area : 220 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {844.800 1223.040 929.280 1300.000} 221 of 225
  VERIFY DRC ...... Sub-Area : 221 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {929.280 1223.040 1013.760 1300.000} 222 of 225
  VERIFY DRC ...... Sub-Area : 222 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1013.760 1223.040 1098.240 1300.000} 223 of 225
  VERIFY DRC ...... Sub-Area : 223 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1098.240 1223.040 1182.720 1300.000} 224 of 225
  VERIFY DRC ...... Sub-Area : 224 complete 0 Viols.
  VERIFY DRC ...... Sub-Area: {1182.720 1223.040 1260.000 1300.000} 225 of 225
  VERIFY DRC ...... Sub-Area : 225 complete 0 Viols.

  Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:01:38  ELAPSED TIME: 0:03:31  MEM: 297.1M) ***
```

Floorplan – check_floorplan

```
Begin Summary
    1 Problem(s) (IMPVFC-98): Net has no global routing and no special routing.
    1 total info(s) created.
End Summary

End Time: Wed Apr 15 16:32:45 2026
Time Elapsed: 0:00:01.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 1 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.1  MEM: 0.000M)
```



1 lỗi connect với io/set_dont_touch_spare (dự phòng) do chưa kết nối với wire nào.

Placement

Zone	% width	Ý nghĩa
INPUT	20%	gần input
CORE	40%	ALU, FSM
ACC	20%	trung gian
STATUS	20%	output

→ X1 + 0.2 * core_w
→ X1 + 0.6 * core_w
→ X1 + 0.8 * core_w

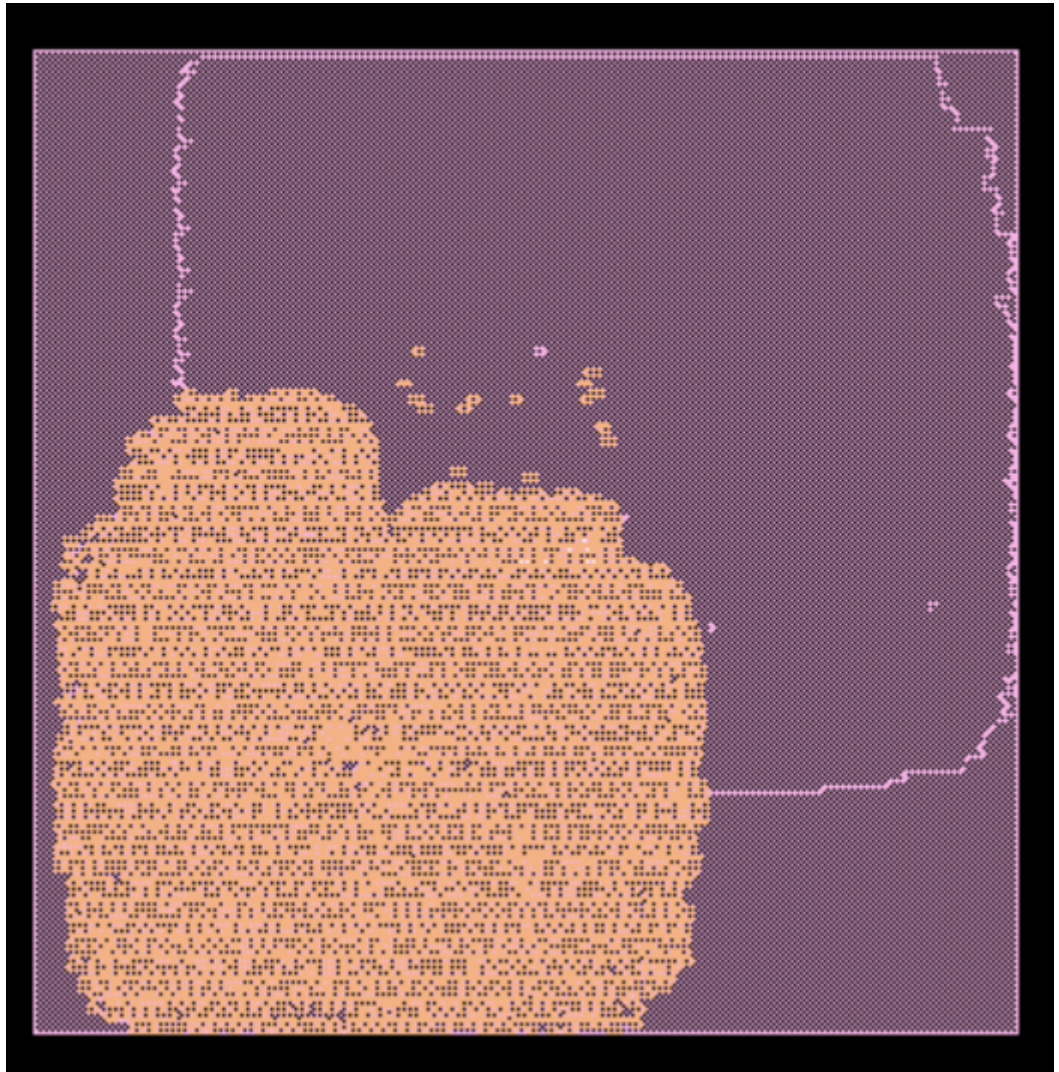
```
@innovus 6> get_db designs .core_bbox
{469.285 489.285 790.715 810.715}
```

Gom ram_wraper và các nhóm input, core, acc, status

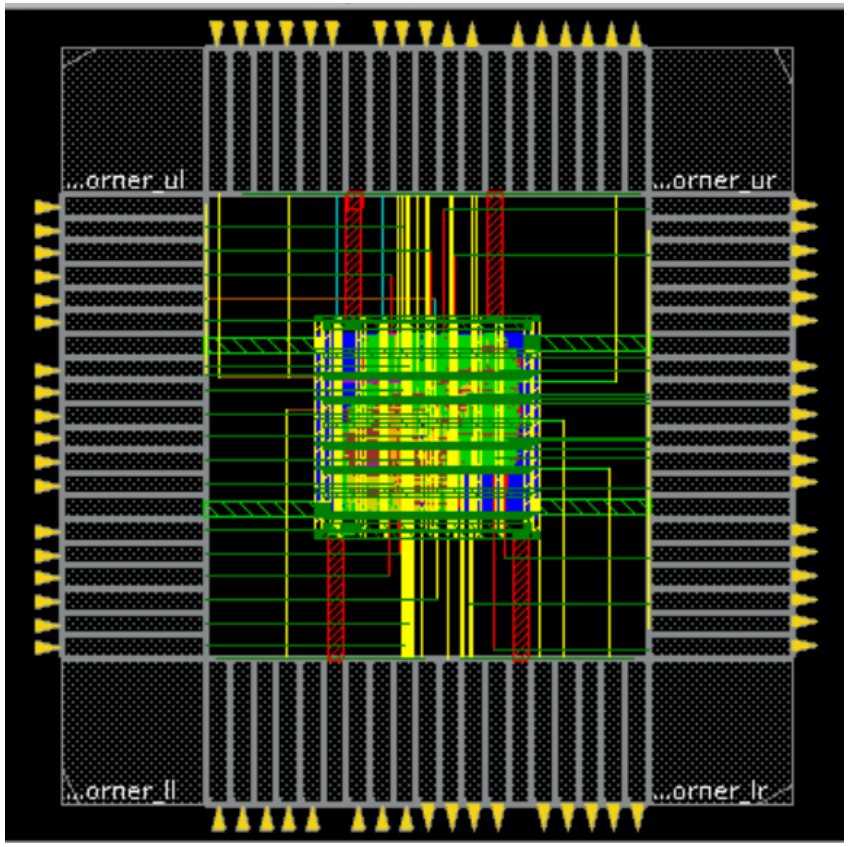
core_w = 790.715 - 469.285 ≈ 321.43
core_h = 810.715 - 489.285 ≈ 321.43

Placement

- Ram_wraper: orange



Placement



Placement – check_timing

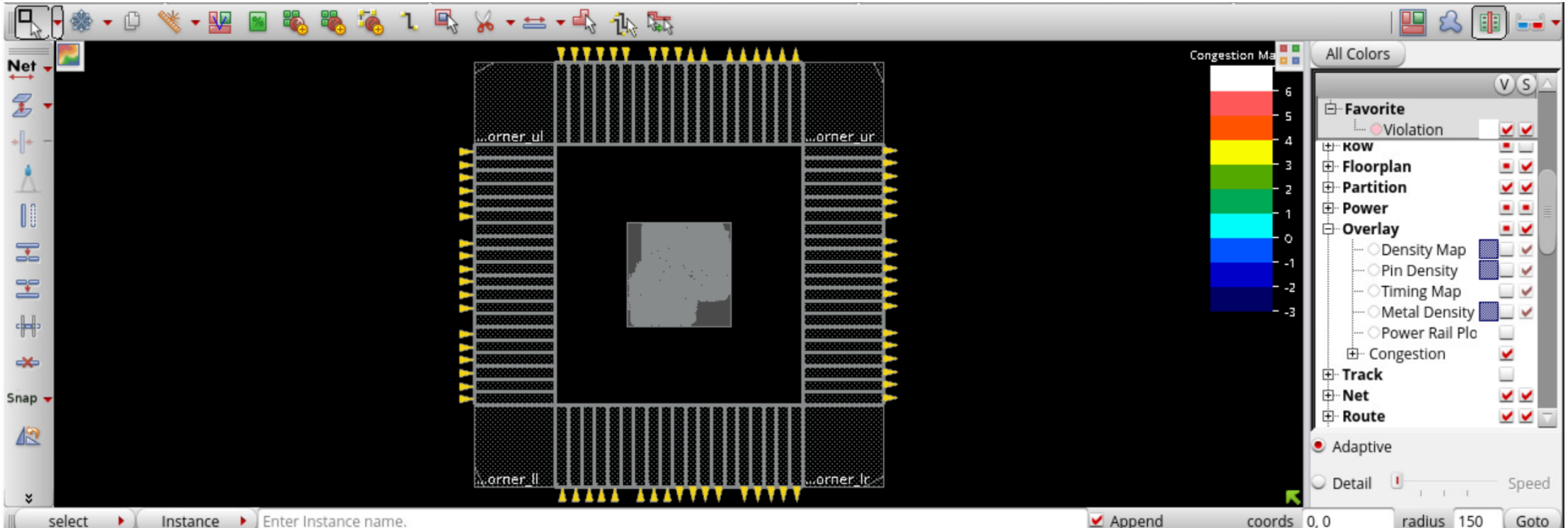
```
"""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""
Path 1: MET (0.122 ns) Setup Check with Pin U_CORE/DP/ZERO_reg/qout_regx0x/CK->SI
               View: default_emulate_view
              Group: reg2reg
         Startpoint: (R) U_CORE/DP/ACC_reg/qout_regx1x/CK
              Clock: (R) clk
           Endpoint: (R) U_CORE/DP/ZERO_reg/qout_regx0x/SI
              Clock: (R) clk

                       Capture       Launch
        Clock Edge:+    2.000        0.000
       Src Latency:+    0.000        0.000
       Net Latency:+    0.000 (I)   -0.050 (I)
           Arrival:=    2.000       -0.050

              Setup:-   0.184
      Required Time:=   1.816
       Launch Clock:=  -0.050
          Data Path:+   1.744
              Slack:=   0.122

#-----------------------------------------------------------------------------
```

Placement – check_timing

```
------------------------------------------------------------------
         time_design Summary
------------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+-----------------------------------------------+
|     ID Number      |                Path Group Name                |
+--------------------+-----------------------------------------------+
|         1          |reg2ckgate                                     |
+--------------------+-----------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.056  | -0.056  |  1.050  |  1.225  |   N/A   |  0.627  |  0.292  |
|           TNS (ns):| -0.288  | -0.288  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|    8    |    8    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4527   |  4267   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 65.654%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
```

Placement – check_congestion

```
@innovus 7> report_congestion -hotspot
[hotspot] +------------+---------------+---------------+
[hotspot] |            |   max hotspot | total hotspot |
[hotspot] +------------+---------------+---------------+
[hotspot] | normalized |          0.00 |          0.00 |
[hotspot] +------------+---------------+---------------+
Local HotSpot Analysis: normalized max congestion hotspot area = 0.00, normalized total congestion hotspot area = 0.00 (area is in unit of 4
 std-cell row bins)
@innovus 8>
```



CTS – check_timing

```
#####################################################################
Path 1: VIOLATED (-0.003 ns) Setup Check with Pin U_CORE/DP/ACC_reg/qout_regx0x/CK->D
               View: default_emulate_view
              Group: reg2reg
         Startpoint: (R) U_CORE/DP/MDR_reg/qout_regx1x/CK
              Clock: (R) clk
           Endpoint: (R) U_CORE/DP/ACC_reg/qout_regx0x/D
              Clock: (R) clk

                       Capture       Launch
         Clock Edge:+   12.000        0.000
        Src Latency:+    0.000        0.000
        Net Latency:+    0.603 (P)    0.356 (P)
            Arrival:=   12.603        0.356

              Setup:-    0.098
        Uncertainty:-    0.100
        Cppr Adjust:+    0.000
      Required Time:=   12.405
       Launch Clock:=    0.356
          Data Path:+   12.053
              Slack:=   -0.003
```

CTS – check_timing

default_emulate_view

ID Number	Path Group Name
1	reg2ckgate

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	1:reg..te
WNS (ns):	-0.158	-0.003	0.866	-0.158	N/A	0.792	0.046
TNS (ns):	-3.170	-0.003	0.000	-3.167	N/A	0.000	0.000
Violating Paths:	34	1	0	33	N/A	0	0
All Paths:	4525	4265	33	35	N/A	91	134

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

CTS – check_timing

```
------------------------------------------------------------------
         time_design Summary
------------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+-------------------------------------------------+
|     ID Number      |                Path Group Name                  |
+--------------------+-------------------------------------------------+
|         1          |reg2ckgate                                       |
+--------------------+-------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.075  |  0.075  |  0.376  |  1.833  |   N/A   |  0.425  |  0.698  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 64.652%
```

Routing – check_drc

ID Number	Path Group Name
1	reg2ckgate

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	1:reg..te
WNS (ns):	-0.138	0.004	0.852	-0.138	N/A	0.753	0.035
TNS (ns):	-2.472	0.000	0.000	-2.472	N/A	0.000	0.000
Violating Paths:	29	0	0	29	N/A	0	0
All Paths:	4525	4265	33	35	N/A	91	134

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Routing – check_drc

```
------------------------------------------------------------------
         time_design Summary
------------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+-----------------------------------------------------+
|     ID Number      |                  Path Group Name                    |
+--------------------+-----------------------------------------------------+
|         1          |reg2ckgate                                           |
+--------------------+-----------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.073  |  0.073  |  0.384  |  1.808  |   N/A   |  0.453  |  0.706  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 64.667%
------------------------------------------------------------------
```

Routing – check_drc

```
 VERIFY DRC ...... Sub-Area: {760.320 1223.040 844.800 1300.000} 220 of 225
 VERIFY DRC ...... Sub-Area : 220 complete 0 Viols.
 VERIFY DRC ...... Sub-Area: {844.800 1223.040 929.280 1300.000} 221 of 225
 VERIFY DRC ...... Sub-Area : 221 complete 0 Viols.
 VERIFY DRC ...... Sub-Area: {929.280 1223.040 1013.760 1300.000} 222 of 225
 VERIFY DRC ...... Sub-Area : 222 complete 0 Viols.
 VERIFY DRC ...... Sub-Area: {1013.760 1223.040 1098.240 1300.000} 223 of 225
 VERIFY DRC ...... Sub-Area : 223 complete 0 Viols.
 VERIFY DRC ...... Sub-Area: {1098.240 1223.040 1182.720 1300.000} 224 of 225
 VERIFY DRC ...... Sub-Area : 224 complete 0 Viols.
 VERIFY DRC ...... Sub-Area: {1182.720 1223.040 1260.000 1300.000} 225 of 225
 VERIFY DRC ...... Sub-Area : 225 complete 0 Viols.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:01:36  ELAPSED TIME: 0:01:41  MEM: 264.1M) ***

1
```

Routing – check connectivity

```
VERIFY_CONNECTIVITY use new engine.

******** Start: VERIFY CONNECTIVITY ********
Start Time: Thu Apr 16 18:33:38 2026

Design Name: cpu_32bit_top
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (1260.0000, 1300.0000)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 18:33:38 **** Processed 5000 nets.
**** 18:33:39 **** Processed 10000 nets.
**** 18:33:39 **** Processed 15000 nets.
**** 18:33:39 **** Processed 20000 nets.

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Thu Apr 16 18:33:39 2026
Time Elapsed: 0:00:01.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:01.0  MEM: 0.000M)
```

Routing – check_antenna

```
@innovus 8> check_antenna
 *** Starting Verify Antenna (MEM: 3214.3) ***

Report File: cpu_32bit_top.verify_antenna.rpt
  VERIFY Antenna ...... Starting Verification
  VERIFY Antenna ...... Using new threading
5000 nets processed: 0 violations
10000 nets processed: 0 violations
15000 nets processed: 0 violations
20000 nets processed: 0 violations
Verification Complete: 0 Violations
 *** End Verify Antenna (CPU: 0:00:02.4  ELAPSED TIME: 3.00  MEM: 0.0M) ***

******* DONE VERIFY ANTENNA ********
1
```